32nd National and 10th International Iranian Conference
ICBME 2025
Tabriz - Iran 19-20 November

Article's Code in ICBME 2025: icbme_1007

Leveraging Online Data to Enhance Medical Knowledge in a Small Persian Language Model

Ghassabi , Mehrdad / Rostami , Pedram / Baradaran Kashani,Hamidreza / Poursina , Amirhossein / Kazemi , Zahra / Tavakoli , Milad

Presenter: Mehrdad Ghassabi
University of Isfahan

ICBME2025

Article's Code in ICBME 2025: icbme_1007

Motivation

- AI and language models are transforming healthcare.
- English models (e.g., MedPalm, ChatDoctor) already provide expert-level medical Q&A.
- But Persian, spoken by millions, still lacks open, reliable medical language models.
- Building such models is crucial for accessible healthcare AI and privacy-friendly local deployment.

Article’s Code in ICBME 2025: icbme_1007

Problem Statement

- No open Persian medical corpus or Q&A dataset existed before.
- The only medical model (Sina-BERT) is closed-source and limited to extractive tasks.
- Small Persian LMs struggle to generate accurate, context-aware medical responses.

Article's Code in ICBME 2025: icbme_1007

English Models

- *ChatDoctor*: fine-tuned LLaMA with 100K QA pairs; strong performance, retrieval-augmented.
- *MedMobile*: optimized for mobile devices; small but efficient.
- *Meerkat*: trained on reasoning from textbooks, closer to human clinical thinking.

Persian Efforts

- *Sina-BERT*: trained on crawled corpus + annotated datasets; but closed-source and encoder-based → unsuitable for generative Q&A.
- Other Persian QA systems (Veisi et al., Darabi): rely on retrieval methodsrather than true generative answers.

As you can see , there are several challenges in this field:

- Datasets, models, and code are not publicly available, blocking reproducibility.
- Focused on extractive solutions; cannot produce fluent, context-aware responses.
- No open benchmark or shared resources → research community left with a "blank slate."

Article's Code in ICBME 2025: icbme_1007

Contributions

- First open-source Persian medical language model — breaking the barrier of closed and inaccessible solutions.
- Built a 90M-token Persian medical corpus from trusted online sources
- Introduced MF3QA, the first free-form Persian doctor–patient Q&A dataset (~20k pairs).
- Created the first Persian medical benchmark by translating MMLU medical tasks + using Iran's IBMSEE exam.
- Demonstrated that a small 8B model can outperform general LMs and pipelines — making secure,
  on-device medical AI possible for Persian speakers.

Article's Code in ICBME 2025: icbme_1007

Dataset Construction

1. Persian Medical Corpus

- ~100k medical articles from trusted Persian sources.
- ~90M tokens → first large-scale Persian medical corpus.
- Covers diseases, treatments, lifestyle; foundation for medical language.

2. MF3QA (Doctor–Patient Q&A)

- Crawled >180k raw pairs from drhast, niniban, doctor-yab, isovisit.
- BFS crawler used on drhast → 120k records in 2 weeks.
- Heavy filtering (short answers) → final ~20k high-quality pairs.
- Train: drhast + niniban | Test: doctor-yab + isovisit + translated K-QA.

3. Benchmarks

- Translated medical portion of MMLU into Persian.
- Added IBMSEE exam questions.
- First reliable benchmarks for Persian medical LMs.

Article's Code in ICBME 2025: icbme_1007

Methodology / Training

Baseline Model

Aya-Expanse-8B chosen for strong Persian support → fluent, grammar-rich output without code-switching.

1. Fine-Tuning (Corpus, 90M tokens)

Used 60% of medical corpus to inject domain knowledge.

Efficiency techniques:

- LoRA (r=8, α=16, dp=5%).
- Gradient checkpointing + accumulation (batch 2 → effective 32).
- FlashAttention-2 for long contexts.

Result: memory-efficient fine-tuning with limited compute.

Article's Code in ICBME 2025: icbme_1007

Methodology / Training

2. Instruction-Tuning (MF3QA, ~20k pairs)

- Applied lighter settings: LoRA (r=2, α=2, dp=0.4), weight decay=0.5.
- Single epoch with Aya-Expanse template.
- Goal: context-aware, empathetic, structured Farsi answers.

3. Efficiency & Sustainability

Training done in ~19h on 1×NVIDIA A100 (40GB).

Carbon footprint ≈ 2.66 kg CO_2e — low-cost and eco-friendly.

Article's Code in ICBME 2025: icbme_1007

Results

- **First Persian Medical LM to pass a national exam**

Achieved 38.69% on the Iranian Basic Medical Sciences Entrance Exam (IBMSEE), above the 36% passing threshold.

- **Superior performance on MMLU (medical tasks, Persian-translated)**

Outperformed Aya-Expanse, Qwen2, and PersianMind across almost all medical sub-domains.
Demonstrated state-of-the-art understanding of Persian medical knowledge.

Article's Code in ICBME 2025: icbme_1007

Results

- **High-quality Free-Form Q&A**

Independent evaluation by GPT-4o showed a clear preference for our model's answers over competing models.
Responses judged as more accurate, detailed, and context-aware.

- **Better than pipeline alternatives**

Our model outperformed translation-based pipelines (Persian → English → Medical LM → Persian).
Faster inference and superior handling of medical terminology, avoiding dangerous mistranslations

Article's Code in ICBME 2025: icbme_1007

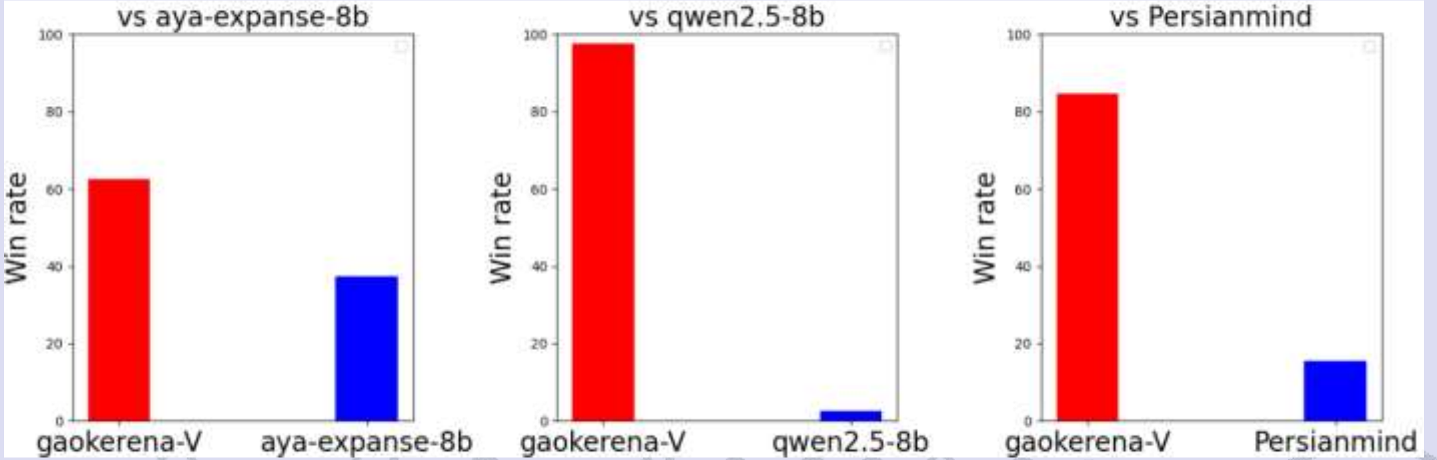
Results

Model	MMLU(avg)	IBMSEE(sep 2023)
Gaokerena(ours)	49.31	38.69
Aya-expanse-8b	46.64	34.52
Qwen 2.5	45.17	33.33
PersianMind	25.89	19.64

Article's Code in ICBME 2025: icbme_1007

Results

Gaokerena-V consistently outperforms Aya-Expanse-8B, Qwen2.5-8B, and PersianMind in free-form Q&A. GPT-4o judges strongly preferring its answers for accuracy, detail, and medical reliability.



Article's Code in ICBME 2025: icbme_1007

Conclusion

- Built the first open-source Persian medical language model.
- Created a 90M-token medical corpus and the MF3QA dataset (~20k Q&A).
- Developed the first Persian medical benchmarks (MMLU translated + IBMSEE).
- Our model passed a national medical exam and outperformed general LMs and pipelines.
- A strong step toward accessible, reliable, and resource-efficient Persian medical AI.

Article's Code in ICBME 2025: icbme_1007

Thanks for your attention

Our data will be available at:

- https://github.com/Mehrdadghassabi/Gaokerena-V
- https://huggingface.co/gaokerena